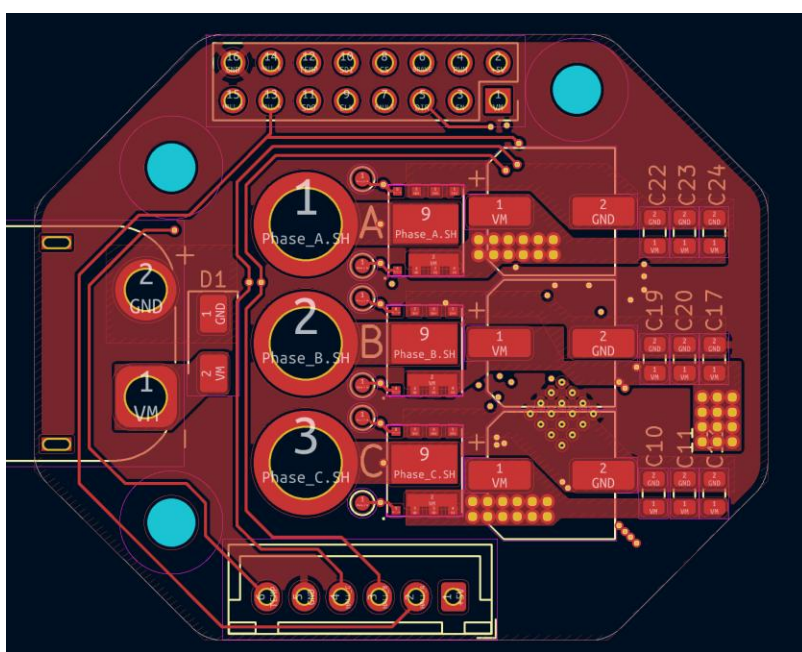
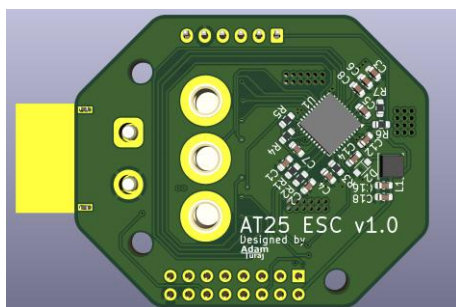
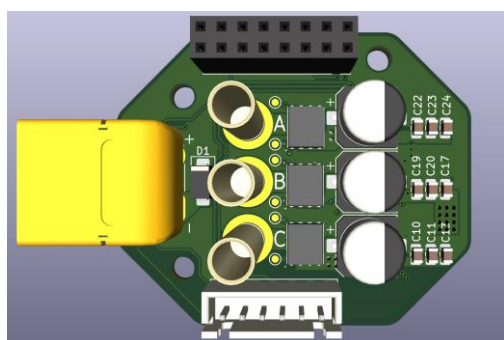
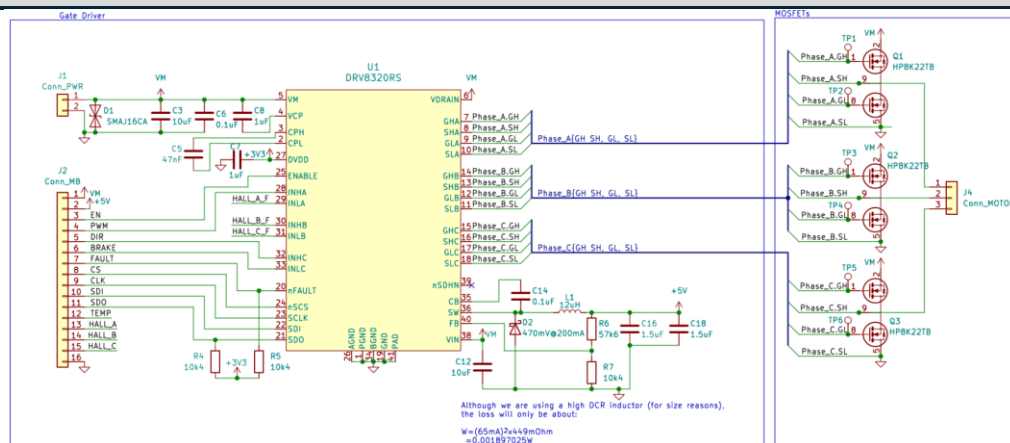


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Gate Driver /  
MOSFET  
portion of the  
schematic



## Sensored BLDC ESC ↗

August 2025 – September 2025

### Task

- Design a motor driver for a 2S sensored BLDC motor. Ensure the board can continuously supply 25A, with peak current capability up to 50A.
- Use the DRV8323 1x PWM control mode to simplify firmware requirements and avoid the need for complex field-oriented control algorithms.

### Process

- Designed the schematic and 4-layer PCB layout in KiCad.
- Routed high-current motor power on a dedicated power layer with wide traces.
- Separated power and signal paths to reduce EMI and improve reliability.
- Added low-pass filters to the Hall-effect sensor inputs for cleaner feedback signals.

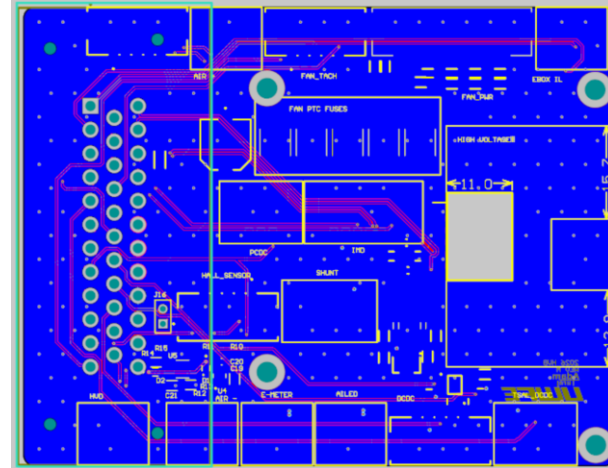
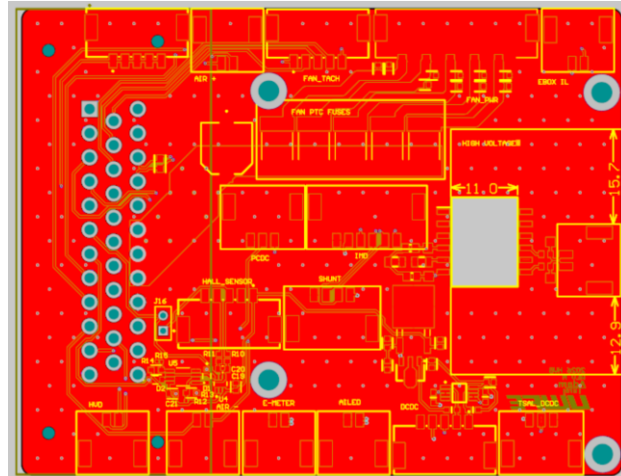
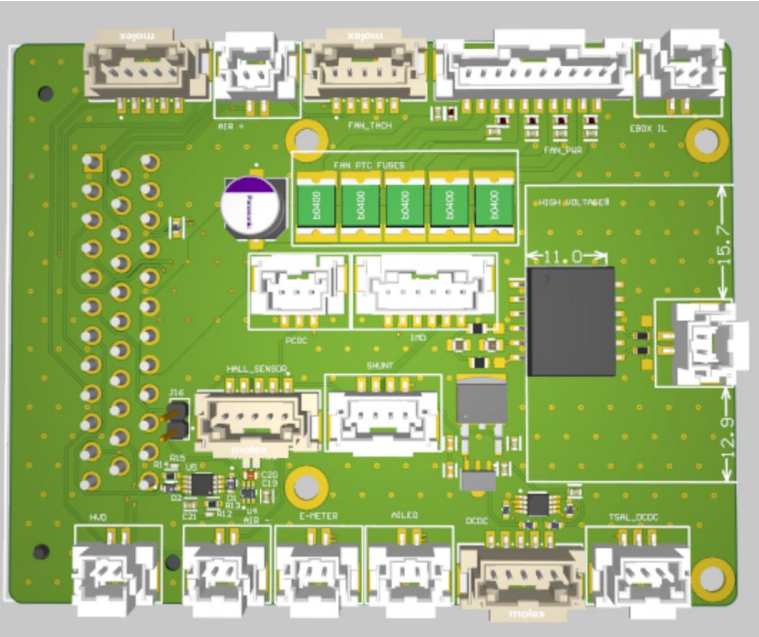
### Result

- Routed all power, sensing, and communication lines within a compact 49 × 39mm PCB.
- Organized the layout for easier assembly, debugging, and understanding.
- Completed a fabrication-ready PCB design that demonstrates compact high-current ESC layout practices.

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## High Voltage Breakout

November 2025 – December 2025

### Task

- Create a PCB to consolidate 16 wire bundles into a single connector.
- Integrate an I2C extender, ISO-SPI transformer, and Hall-effect amplification circuit.
- Keep the PCB compact, FSAE-compliant, and easy to mount within the HV system.

### Process

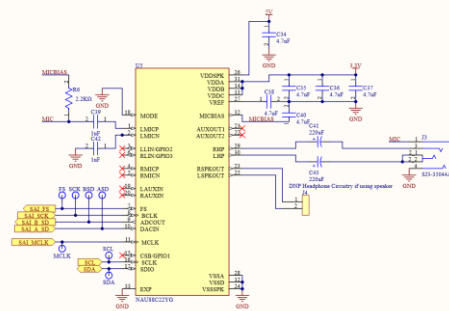
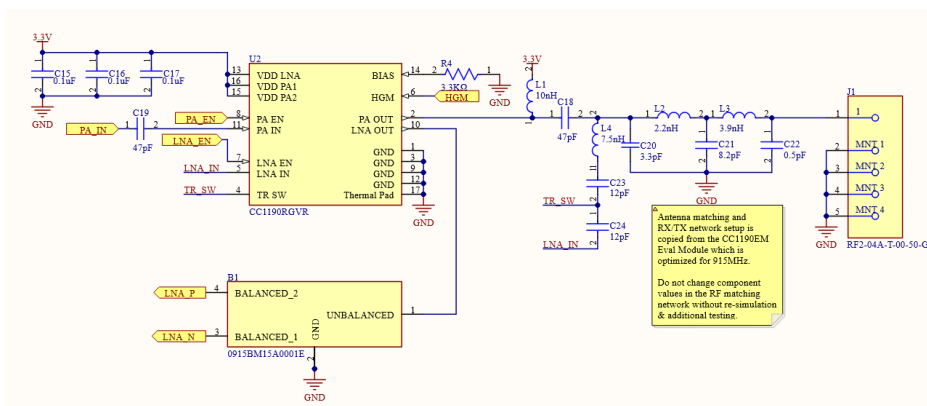
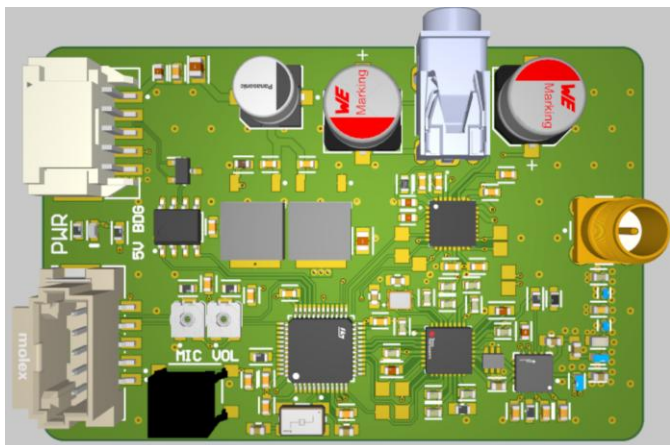
- Developed the 4-layer PCB in Altium.
- Spaced connectors for easier soldering, plugging, and serviceability.
- Added PTC protection for fan outputs to prevent overcurrent without the need to physically reset fuses.
- Included clear labeling, connector spacing, and a cutout for ISO-SPI isolation.

### Result

- Installed the HVB in the battery pack, where it performed reliably.
- Central AMPSEAL connector placement simplified mounting and cable access.
- A second revision added an ISO-SPI bypass to place the transformer closer to the destination.

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# Digital Driver Radio

## October 2025 – Present

# Task

- Design an integrated driver radio with half-duplex digital communication.
- Use a digital protocol to improve audio quality and reliability.
- Add CAN communication for wireless telemetry and data logging.

# Process

- Created a compact 4-layer PCB in Altium within a  $65 \times 43\text{mm}$  footprint.
- Used the TI CC1120 for 915 MHz communication and a 24-bit audio CODEC for microphone and speaker audio.
- Implemented the CC1190 RF front-end with datasheet-based matching and filtering circuits.
- Designed the board to support both driver and pit crew configurations.

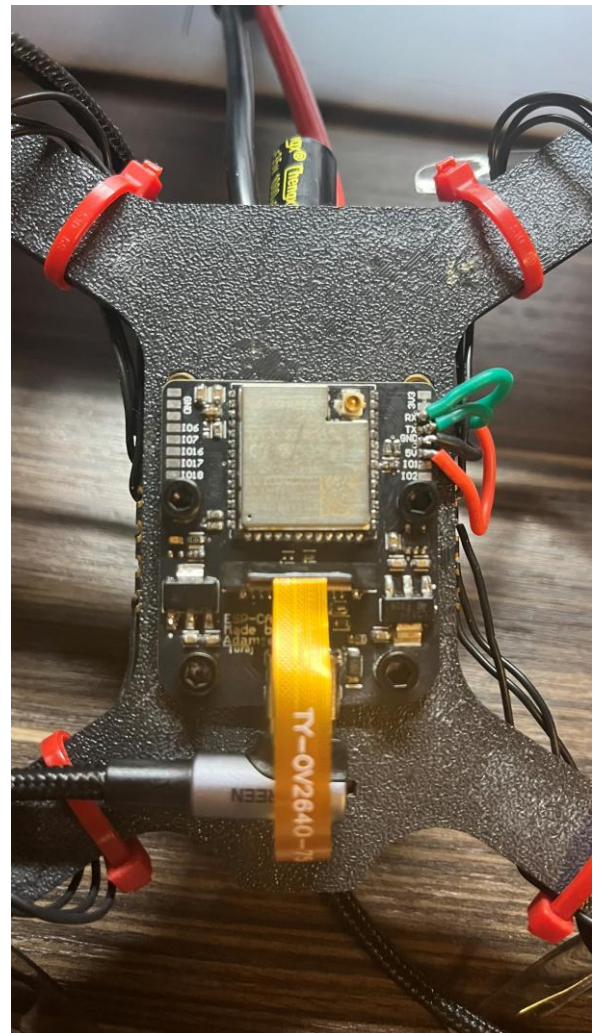
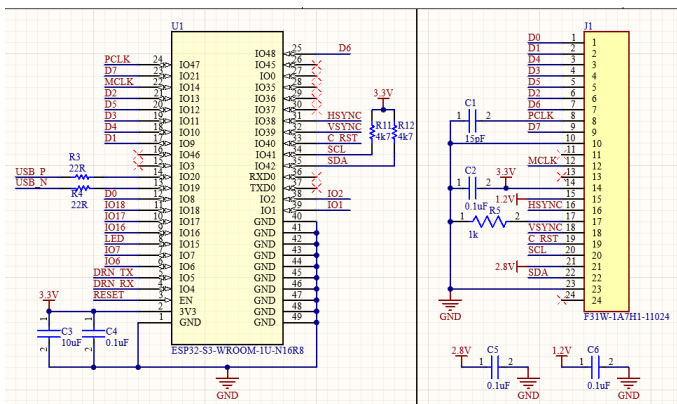
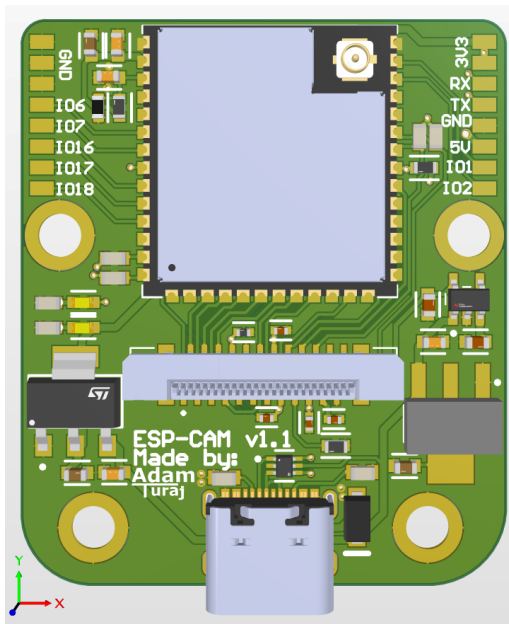
## Result

- Assembled the board and began testing core functionality.
- Successfully played audio through the headphone output, with microphone input testing still in progress.
- Verified SPI communication with the CC1120, with full RF validation pending access to proper test equipment.

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## ESP32 Camera<sup>3</sup>

March 2026 – Present

### Task

- Design a compact PCB integrating an ESP32 wireless module with a camera interface.
- Break out available GPIO pins for future expansion and debugging.

### Process

- Traced a simple 2-layer PCB in Altium to fit flat under the drone.
- Prioritized short, continuous camera data traces for reliable signal routing.
- Added voltage regulation to properly supply the camera module.
- Selected decoupling capacitors to improve camera clock and signal stability.

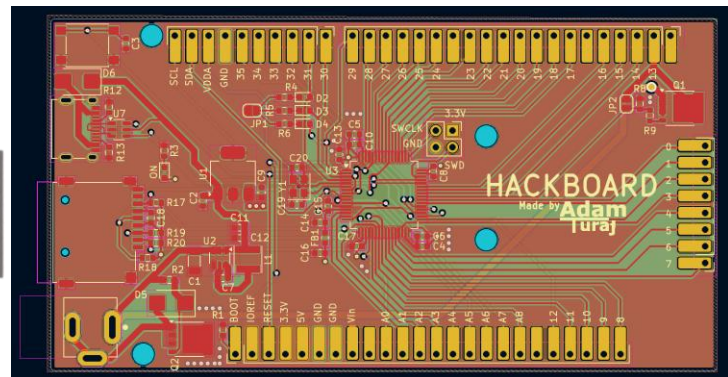
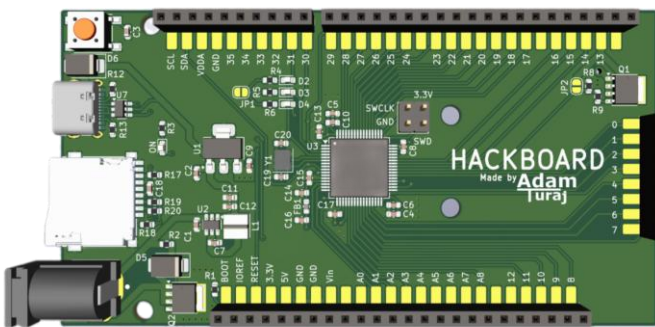
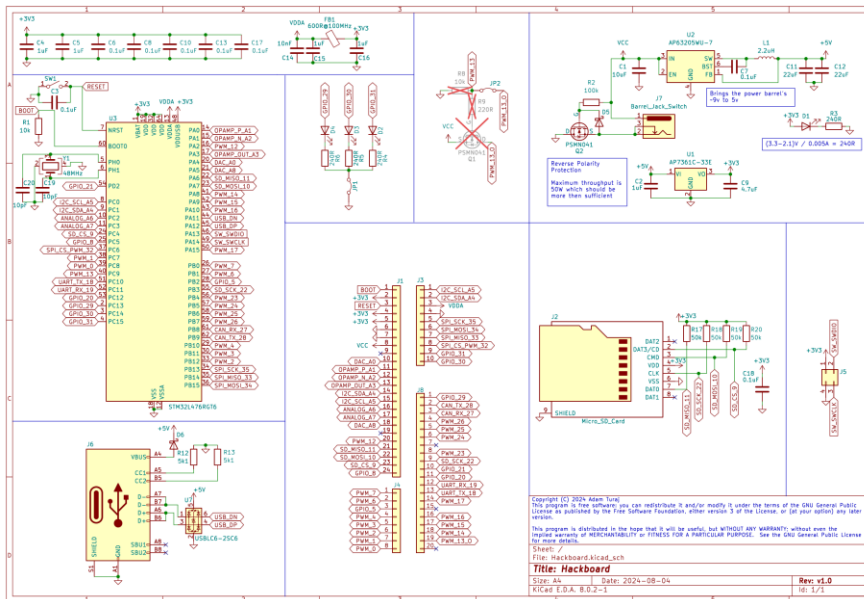
### Result

- Successful flashing and wireless testing of ESP32 and Bluetooth.
- Achieved communication with flight control unit over UART.
- Camera testing still a work in progress.

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## Hackboard ↗

August 2024 – September 2024

### Task

- Create an all-in-one Arduino-compatible development board that improves common usability issues.
- Match Arduino Uno header and screw-hole placement for shield compatibility.

### Process

- Designed a compact 4-layer PCB in KiCad.
- Used a low-power MCU and efficient buck converter to improve battery life.
- Added extended header pads, addressable LEDs, an SD card slot, and a 54W PWM motor driver.
- Matched Arduino Uno headers and mounting holes for shield support.

### Result

- Completed a clean PCB layout with proper trace spacing to reduce EMI.
- Used wide traces for the motor driver and differential routing for USB signals.
- Organized the schematic so the board is easy to understand, debug, and modify.